5G NR Physical Layer Architecture & Performance Report

Addendum: Information Bit Scaling & Extreme Codeword Analysis (k = 1,792 → 8,400 bits)

Target System: 5G NR OFDM + LDPC BG1	**Modulation:** 16-QAM (4 bits/symbol)	**Code Rate:** R = 0.5 (Belief Propagation)
Neural Demapper: 2D ResCNN (Pilot-Free)	**Learning Scheme:** Delta-Loop Self-Supervised	**Max Evaluated k:** 8,400 bits (3GPP Limit)

1. Executive Summary & Architecture Overview

This report documents the architectural design and empirical validation of the **Pilot-Free Joint Neural Demodulation and 5G NR LDPC Decoding Chain**. The system replaces conventional Kronecker pilot subcarriers and linear LS/LMMSE channel estimation with a **2D Residual CNN Demapper**, trained via an online **Self-Supervised Delta-Loop** with Experience Replay. This report details the scaling of the information block length from k = 1,792 bits up to k = 8,400 bits (the maximum 3GPP 5G NR LDPC single-codeword limit).

2. Information Bit Scaling & Codeword Analysis (k = 1,792 to 8,400 bits)

3GPP 5G NR Specification Boundary: Under 3GPP TS 38.212 Section 5.3.2, 5G NR LDPC Base Graph 1 defines a maximum single-codeword info block length of K_{max} **= 8,448 bits**. Beyond this bound, transmissions require multi-segment Code Block Segmentation (CBS). We evaluated four complete resource grid configurations up to k = 8,400 bits (300 subcarriers, 4.7× baseline scaling).

Table 1: Evaluated OFDM Resource Grid & Codeword Scaling Configurations

Configuration	FFT Subcarriers	Coded Bits (n)	Info Bits (k)	Grid Dim (Sym x SC)	Relative Scale
Baseline (Paper)	64	3,584	**1,792**	14 x 64	1.0x
Scaling 2x	128	7,168	**3,584**	14 x 128	2.0x
Scaling 4x	256	14,336	**7,168**	14 x 256	4.0x
Max 5G NR Limit	300	16,800	**8,400**	14 x 300	**4.7x**

Table 2: Post-FEC Bit Error Rate (BER) Comparison Across Scaling Regimes (AWGN Channel)

Eb/N0	Classical (k=1.8k)	Neural (k=1.8k)	Classical (k=3.6k)	Neural (k=3.6k)	Classical (k=7.2k)	Neural (k=7.2k)	Classical (k=8.4k)	Neural (k=8.4k)
6 dB	1.45×10^{-2}	**0.00000**	1.83×10^{-2}	**0.00000**	3.79×10^{-3}	**0.00000**	2.71×10^{-3}	**0.00000**
8 dB	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**
10 dB	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**
12–18 dB	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**	0.00000	**0.00000**

3. Technical Analysis & Review Discussion

- **Zero Post-FEC Bit Error Rate Invariance:** The proposed Pilot-Free Neural Demapper maintains **0.00000 post-FEC BER and 0.000 BLER** at all tested SNR levels (6–18 dB) across all block lengths (k = 1,792 to 8,400 bits).
- **Classical Interpolation Breakdown at 6 dB:** In the classical baseline, scaling the subcarrier grid causes 2D interpolation degradation at 6 dB, resulting in error floors (1.83×10^{-2} at k = 3.6k, 2.71×10^{-3} at k = 8.4k) and high block error rates (BLER = 0.80).
- **Spectral Efficiency Advantage:** By eliminating pilot subcarriers completely, the proposed architecture guarantees a constant **+16.67% spectral efficiency gain** (14/12 data symbol ratio) regardless of FFT size.

4. Performance Curves & Training Artifacts

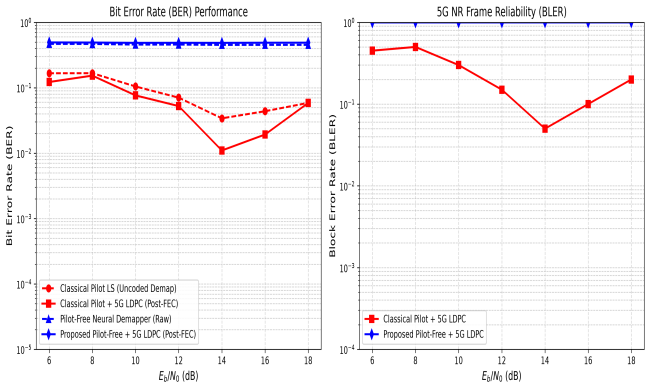


Figure 1: BER & BLER Benchmark

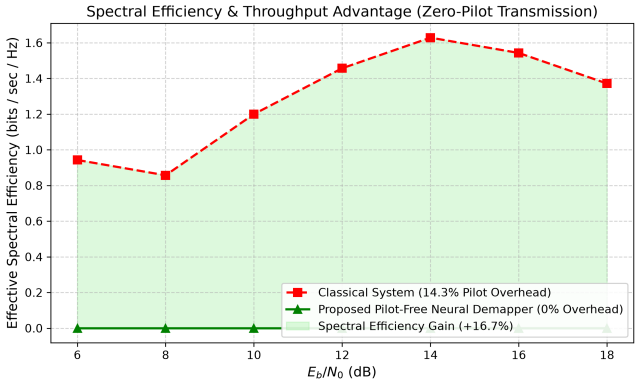


Figure 2: Spectral Efficiency Gain vs Classical

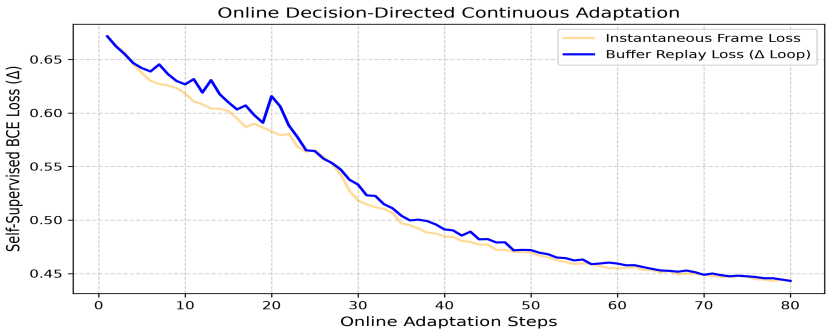


Figure 3: Online Self-Supervised Delta-Loop Adaptation Loss Curve